

Advanced Population Modeling : Integrated Analysis, Diagnostics

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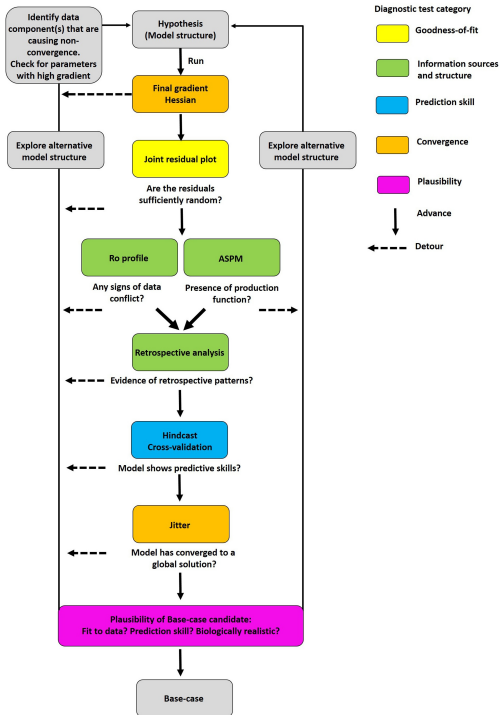
Integrated Analysis

- ▶ Often have complicated population model components, but can fit to a range of data conditions, from data-limited to data-rich
 - ▶ Can fit to disaggregated directly
 - ▶ May be able to match system mechanisms and data generating processes more closely (e.g. model length-based processes as such, ie selectivity, retention, mortality, etc.
- ▶ Can allow for direct inclusion of range of population drivers, environmental factors, etc.
- ▶ Statistical estimation is more challenging
 - ▶ thousands(!) of estimated parameters
 - ▶ many likelihood components

“If applying a VPA is like driving a car, then developing an integrated model is like flying a helicopter”

Integrated Analysis & diagnostics

- ▶ OK, you fit your model. Now what?
- ▶ Just as with our classical statistical models (e.g. regression, etc.), we need to check and evaluate our models - this becomes even more challenging in an integrated analysis framework.
- ▶ Gradient, Parameter checks
- ▶ Goodness of fit
- ▶ Plot data & model predictions
- ▶ Residuals
- ▶ Plot model results
- ▶ Other diagnostics (e.g. retrospective patterns)



Checks

- ▶ Check for convergence
- ▶ Gradient (& components) close to zero?
- ▶ Are parameters running to bounds?
- ▶ Infinite/NaN variance estimates
- ▶ Are parameters correlated?

Goodness of fit

- ▶ Does the model predict the data well?
- ▶ For age/length data, first look at general trend, then individual years/areas, etc.
- ▶ Breakout model fit by likelihood components
- ▶ Starting with a simpler model can help ensure you are guiding things in the right direction.
- ▶ e.g. get 'scale' and 'trend' right before turning on time-varying quantities. . .

Look at residuals in statistical space

- ▶ can better see mis-fits
- ▶ Include the observation error estimates on data when inspecting model predictions
- ▶ Composition data - consider One Step Ahead (prediction) residuals

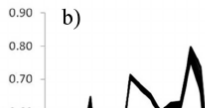
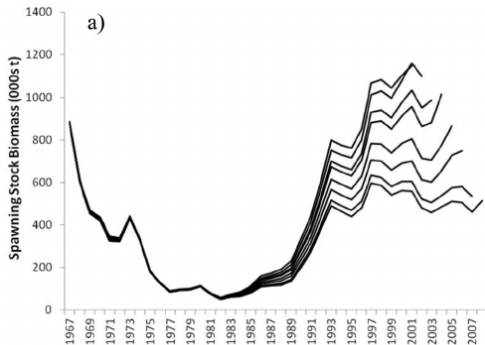
Sensitivity analysis

- ▶ Stock assessments often arrive at a 'base-case'.
- ▶ choice of data, model structure, etc.
- ▶ sensitivity analyses demonstrate change in fit/predictions associated with other hypotheses.
- ▶ can be used to show robustness to uncertainties.

Retrospective patterns

- ▶ may be dubbed 'error' or 'bias'.
- ▶ how model predictions change as data points are removed from the time series

(Brooks & Deroba 2015)



Conflicting data

- ▶ How to predict given some data (data points or entire datasets) that models don't explain well?

Possible causes:

- ▶ True outliers
- ▶ Mis-specified models (i.e., they contain incorrect assumptions, typically over-simplifications)

Solutions:

- ▶ A more complex model with increased flexibility
- ▶ Throw out some of the data
- ▶ Put less weight on data that are considered to be a misrepresentation?

Composition data vs abundance

- ▶ Common to want to reconcile information on the age composition and the total numbers (abundance indices).
- ▶ Francis (2013) suggests giving less weight to the age composition data than the abundance data (“right-weighting” rather than “down-weighting”).
- ▶ We have a limited understanding of the statistical characteristics of the data arising from how we collect fish.
- ▶ Under our statistical assumptions, ideally residual variance should approximately equal the input observation error variance. This is sometimes used as a guidance for how much to weight data (but see recent papers on this topic!)

Abundance indices

- ▶ Variance estimates from sampling design or model standard errors are likely under-estimates of true variance.
- ▶ Consider variance inflation:

$$CV_{use} = \sqrt{CV_t^2 + \sigma_{infl}^2}$$

* ideally, residual variance should approximately equal the input observation error variance.

Composition data

- ▶ 1 fish measured $\neq$ 1 multinomial sample.
- ▶ iterative reweighting
- ▶ multiple methods
- ▶ much discussion, some good practices emerging

Environmental drivers / variables

- ▶ Fair to say that how to 'properly' treat these data has received less attention in the fisheries literature until recently
- ▶ Though methods have existed for long time and are well developed
- ▶ Integrating mechanistic relationships between environmental data and population parameters well established in statistical ecology (e.g. population models for terrestrial animals)
- ▶ 3 general methods:
 - ▶ covariate approach
 - ▶ environment as data
 - ▶ latent variable

Ensemble modeling

- ▶ Different way of dealing with choosing the wrong model.
- ▶ Interpret output of multiple models and incorporating this into the one prediction to rule them all.
- ▶ Reasonably successful in fields like hurricane prediction.
- ▶ But, how do we combine model predictions? Which model predictions get to be part of the ensemble?

Recommendations (via Lewis Barnett)

- ▶ Use reasoning outlined by Maunder & Piner when model diagnostics indicate that you have a fitting problem.
- ▶ If you decide to use data weighting, do not use McAllister and Ianelli (1997) method based on the arithmetic mean.
- ▶ Instead, use Francis' method TA1.8 (Francis 2011) because it accounts for correlations among data, or use an extension of McAllister and Ianelli employing the harmonic mean.
- ▶ NB: data weighting will not account for biases due to model misspecification, so consider other solutions.
- ▶ Consider using a self-weighting statistical model for the observations
- ▶ Use OSA residuals
- ▶ Thorson & Nielsen (2016) propose using mixed-effects models with time-varying “nuisance” variables like recruitment, selectivity, growth or natural mortality modeled as a random effect.